

Relation Algebra

§6. Recursive Programs

§6.1. Digits of a number

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$.
 $(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

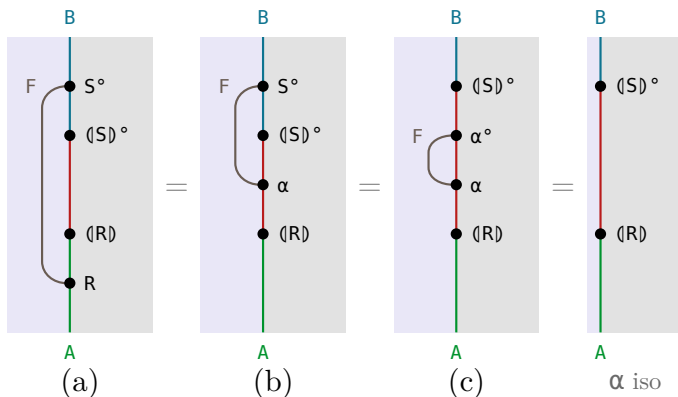
§6.3. Hylomorphisms

$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X) \mathbb{R}) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

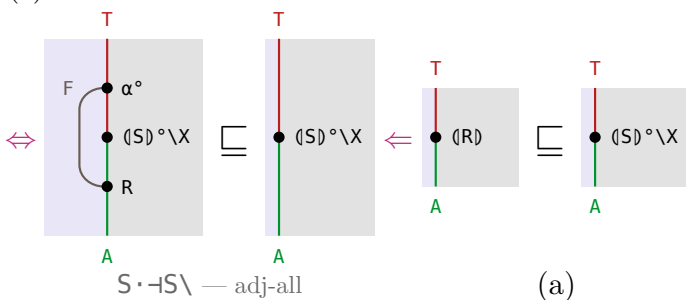
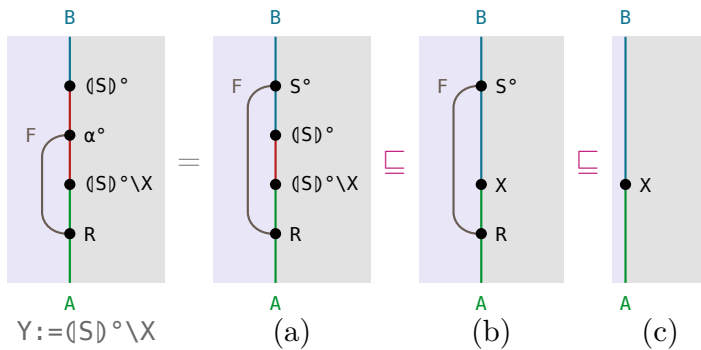
$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R})) \mathbb{R} = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term $\mathbb{S}^\circ$ represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and $\mathbb{R}$ the recombination stage; $\mathbb{R} : \mathbb{F}\mathbb{A} \rightarrow \mathbb{A}$, $\mathbb{S} : \mathbb{F}\mathbb{B} \rightarrow \mathbb{B}$, $\alpha : \mathbb{F}\mathbb{T} \rightarrow \mathbb{T}$ initial



$$\mathbb{S}^\circ F(X) \mathbb{R} \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation

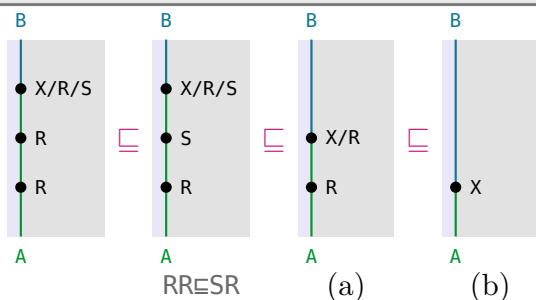


§6.4. Fast exponentiation and modulus computation

§6.5. Unique fixed points

$$\text{inductive}(\mathbb{S}) \wedge \mathbb{R} \sqsubseteq \mathbb{S} \mathbb{R} \implies \text{inductive}(\mathbb{R}) \quad (6.5a)$$

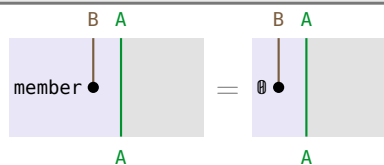
$\mathbb{R}$ is inductive when $\mathbb{R} \sqsubseteq \mathbb{S} \mathbb{R}$ for an inductive $\mathbb{S}$



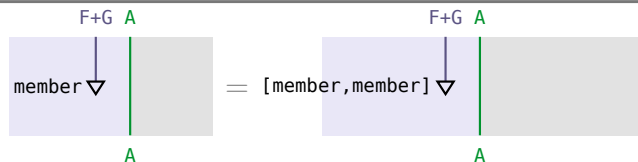
$$\text{member} = \emptyset$$

(6.5b)

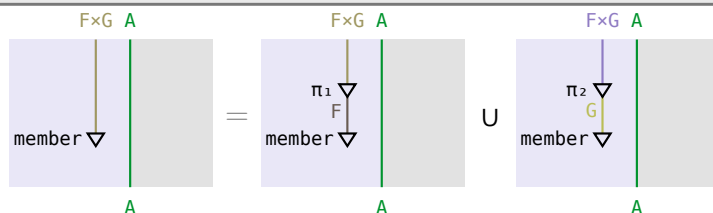
a constant relator records no elements, so its membership is empty



$$\text{member} = [\text{member}, \text{member}]$$

 a member of $F+G$ is a member of whichever summand the value is in


$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

 a member of $F \times G$ is a member of either component


$$\text{inductive}(\text{Smember}) \wedge X = \text{SF}(X)R \wedge Y = \text{SF}(Y)R \Rightarrow X = Y$$

(6.5c)

 when S member is inductive, two solutions X, Y of $X = \text{SF}(X)R$ are equal

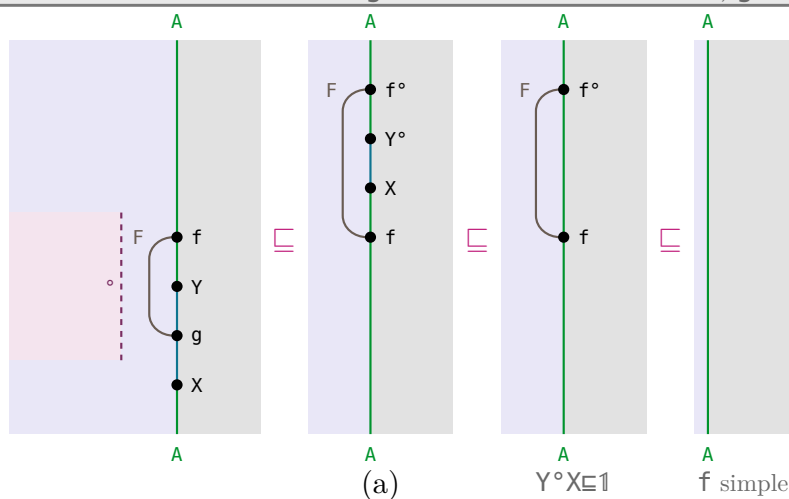
$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(\text{S}) \wedge \text{Entire}(\text{R}) \wedge \text{SF}(X)R \subseteq X \Rightarrow \text{Entire}(X)$$

(6.5d)

 when S member is inductive and S, R are entire, a pre-fixed point of $X \mapsto \text{SF}(X)R$ is entire

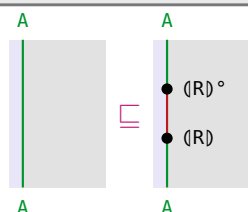
$$\text{inductive}(\text{gmember}) \wedge \text{Map}(\text{g}) \wedge \text{Map}(\text{f}) \wedge X = \text{gF}(X)\text{f} \Rightarrow \text{Map}(X)$$

(6.5e)

 when g member is inductive and f, g are maps, the solution of $X = \text{gF}(X)\text{f}$ is a map

 (a) $X = \text{gF}(X)\text{f}$, g simple

$$\text{inductive}(\text{R}^\circ \text{member}) \wedge 1 \in \text{R}^\circ \text{R} \Rightarrow 1 \in (\text{R})^\circ (\text{R})$$

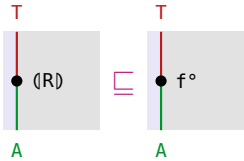
(6.5f)

 when $\text{R}^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective

 (a) (6.5d) at $\text{S} = \text{R}^\circ$

(6.5g)

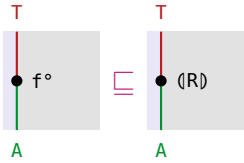
$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \wedge \text{inductive}(\alpha^\circ \text{member}) \Rightarrow f^\circ = \langle R \rangle$

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)

(a) shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$



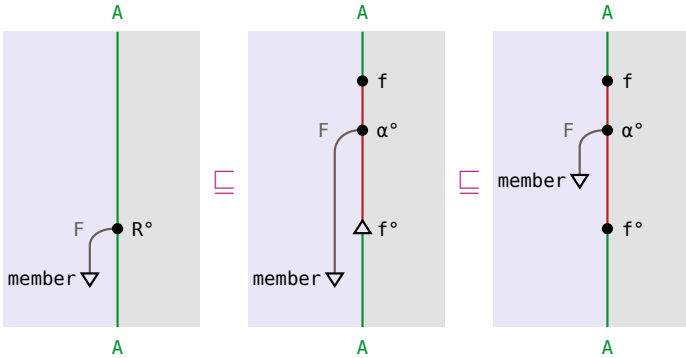
(a)

(a) (6.5f) and $\langle R \rangle f \sqsubseteq 1$

(6.5h)

$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member} f^\circ$

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



(a)

(b)

(a) $Rf \sqsubseteq F(f)\alpha$, shunting

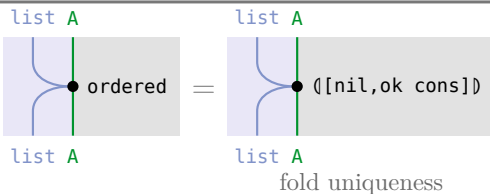
(b) member lax natural

§6.6. Sorting by selection

(6.6a)

$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

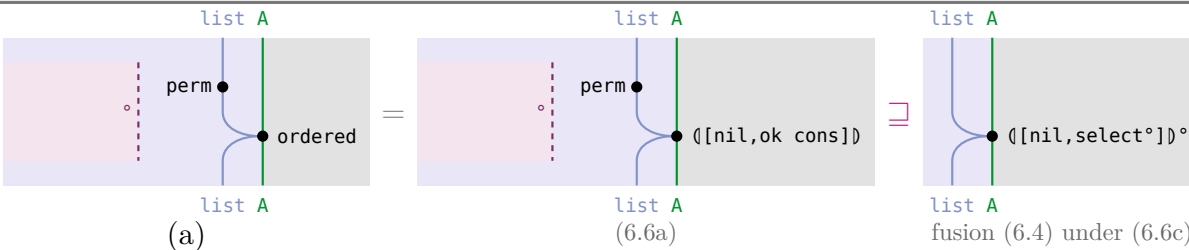


fold uniqueness

(6.6b)

$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$

every output of unfolding the input by **select** is a sorted permutation of it



(a)

(6.6a)

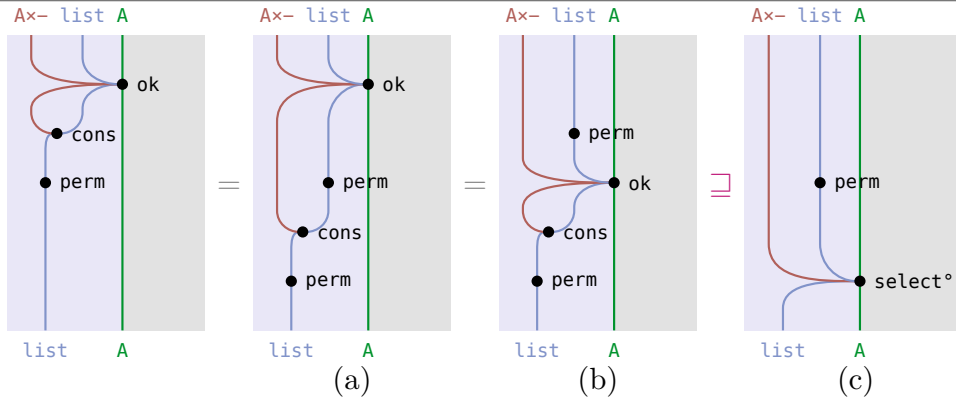
fusion (6.4) under (6.6c)

(a)

$\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

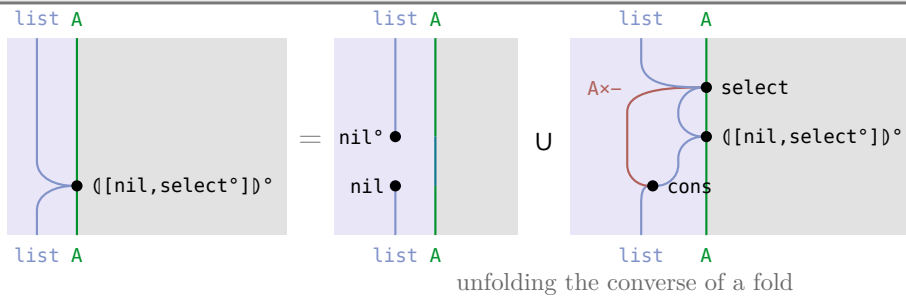
$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** cones of a permutation



$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

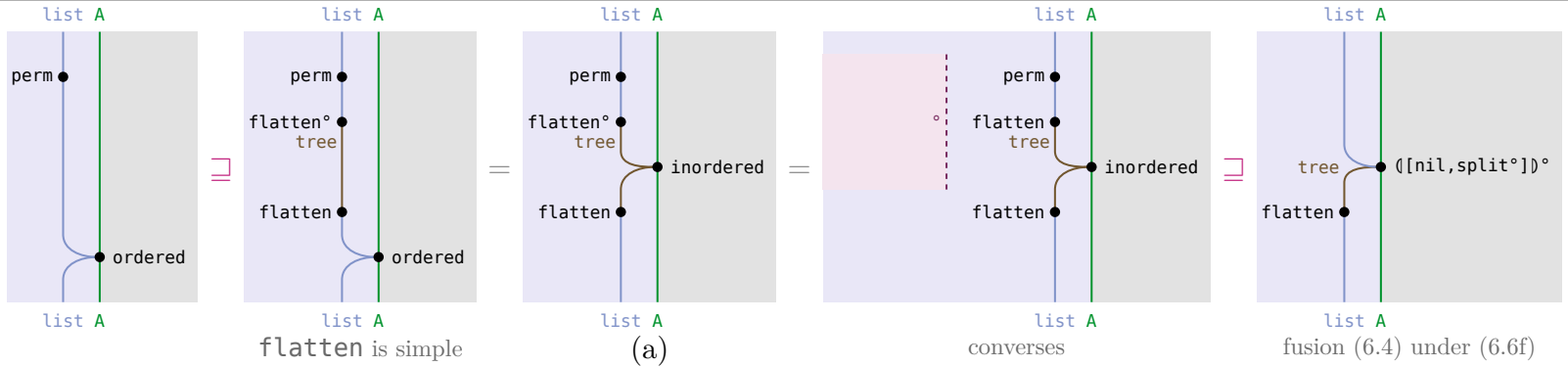
selection sort returns $[]$ on $[]$, and otherwise selects (a, y) , sorts y and conses a back on



$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

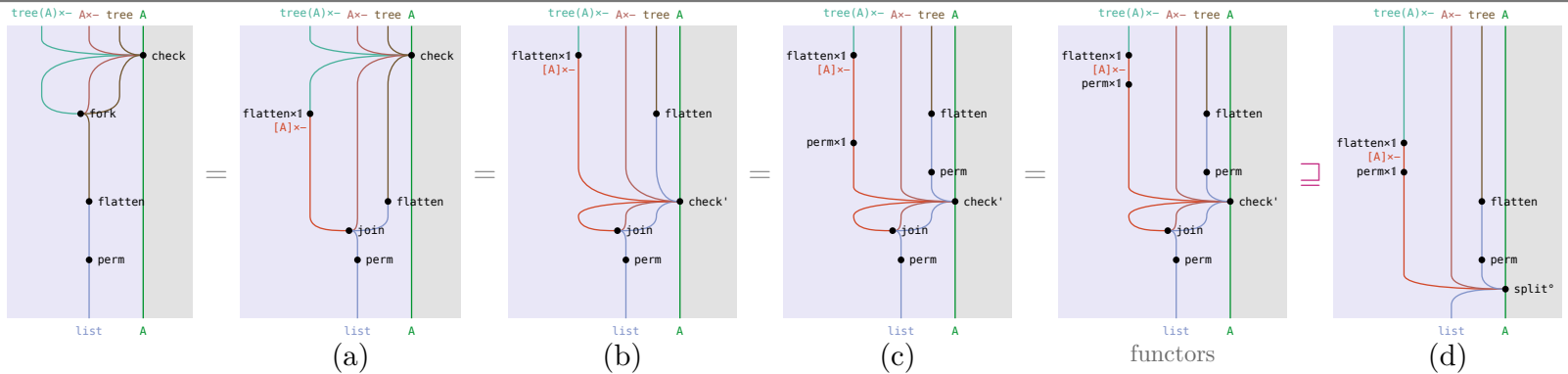
$$([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered}$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check}' \text{ fork flatten perm} \quad (6.6f)$$

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

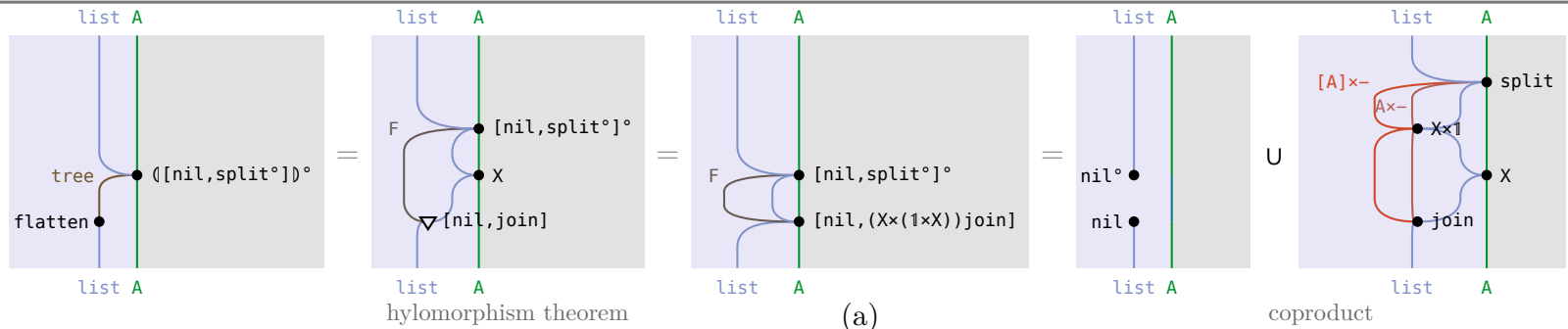


$$([nil, split^\circ])^\circ \circ flatten = nil^\circ nil \cup split$$

(6.6g)

$$((([nil, split^\circ])^\circ \circ flatten) \times (1 \times ([nil, split^\circ])^\circ \circ flatten)) \circ join$$

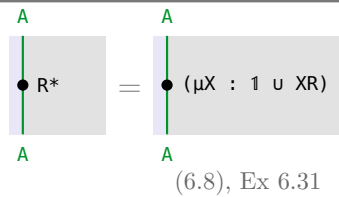
quicksort returns [] on [], and otherwise splits into (x, a, y), sorts x and y and joins them around a



§6.7. Closure

$$R^* = (\mu X : 1 \cup XR)$$

(6.7a)

 closing R by composing it on the left and closing it by composing on the right give the same relation R^*


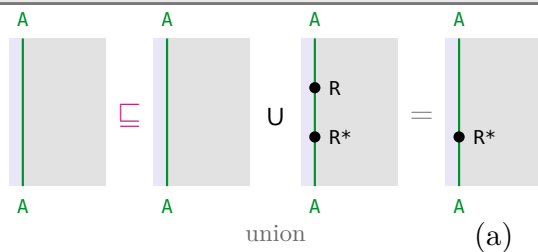
$$1 \in X \wedge XX \in X \implies R \in X \iff R^* \in X$$

(6.7b)

 for a preorder X , X contains R exactly when it contains R^*

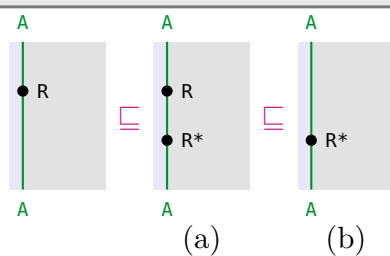
$$1 \in R^*$$

(6.7c)

 R^* is reflexive


$$R \in R^*$$

(6.7d)

 R^* contains R


(a) $1 \in R^*$ — (6.7c)
(b) fixed point

(6.7e)

$R^*R^* \sqsubseteq R^*$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

(a) composition distributes over $\cup$
 (b) $RR^* \sqsubseteq R^*$ — (6.7d)

(6.7f)

$1 \sqsubseteq X \wedge XX \sqsubseteq X \wedge R \sqsubseteq X \implies R^* \sqsubseteq X$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

$1 \sqsubseteq X, R \sqsubseteq X$
 $XX \sqsubseteq X$

(6.7g)

$\text{inductive}(R) \wedge X = 1 \cup RX \implies X = R^*$

when R admits induction, R^* is the only X equal to $1 \cup RX$

(a) $(R^* = R^*/X) \cap R^* \sqsubseteq R^*/X$

(6.7h)

$\frac{R^*}{3} = (\frac{1}{3}, \frac{RR^*}{3}) \text{ cup}$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors

(a) $\wedge(R \cup S) = (\wedge(R), \wedge(S)) \text{ cup}$

(6.7i)

$R - \emptyset = R$

taking nothing away leaves R

$R \cup (S - R) = R \cup S$

beside R , only the part of S outside R adds anything

$R - (S \cup T) = R - S - T$

taking away S and then T takes away $S \cup T$

$(R \cup S) - T = (R - T) \cup (S - T)$

taking T away from a union takes it away from each part

(6.7j)

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$

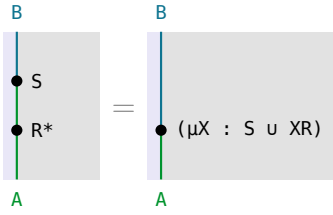
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

Ex 6.35

$$SR^* = (\mu X : S \cup XR)$$

(6.7k)

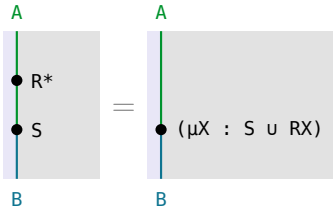
S followed by any number of R steps is the least X containing S and closed under a further R step



p.160

$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front

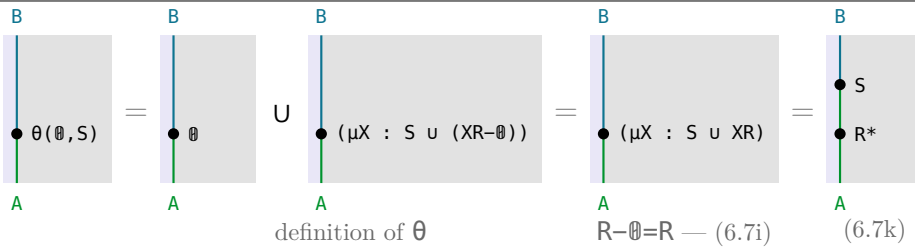


Ex 6.32

$$\theta(\emptyset, S) = SR^*$$

(6.7l)

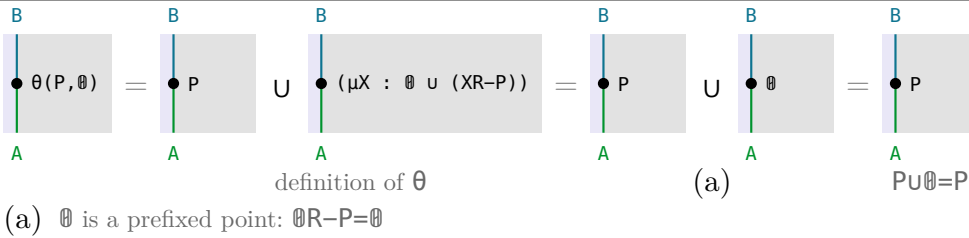
started with nothing found and S to explore, θ computes SR^*



$$\theta(P, \emptyset) = P$$

(6.7m)

with nothing left to explore, θ returns what it has found



(a) $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$

$$\theta(P, Q) = \theta(P \cup Q, QR - P - Q)$$

(6.7n)

one round moves Q into the found part and explores the new R -successors of Q that are in neither

